

Factory Reallocation & Shipping Optimization for Nassau Candy Distributor

*A Machine Learning Approach to Predictive Lead-Time Estimation and Scenario-Based
Resource Allocation*

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This paper presents a demonstration project developed during an internship at Unified Mentor Pvt. Ltd. Data used is simulated for illustrative purposes and clearly labeled as such.

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Disclaimer: This project was developed as part of the Unified Mentor Internship Program. The analysis, predictive models, and dashboard concepts presented herein are for educational and operational-planning demonstration purposes. The views and data interpretations expressed in this report are those of the author and do not necessarily reflect the official policies, internal data structures, or operational practices of Nassau Candy Distributor or any of its affiliates. The dataset referenced throughout this paper is simulated to illustrate methodology and does not represent actual proprietary business records.

Dedication

This project is dedicated to the operations, logistics, and supply chain teams whose daily decisions keep products moving efficiently from factory to shelf.

It is my hope that the predictive intelligence demonstrated in this work contributes to faster, fairer, and more resilient distribution networks — and adds a small piece of evidence to the broader case for data-driven decision-making in operations management.

Abstract

Supply chain inefficiencies in consumer goods distribution often stem from static, rule-based product-to-factory assignments that do not adapt as order patterns, regions, and capacity shift over time. This paper presents a decision-support system developed for Nassau Candy Distributor to evaluate factory reallocation and shipping logistics using predictive machine learning. Using order-level data, we trained Random Forest and Gradient Boosting regressors to estimate shipping lead time as a function of product division, ship mode, region, and distance to factory. The trained model powers a Scenario Simulation Engine that compares a product's current factory assignment against alternative factories and estimates the resulting change in lead time and gross profit. The system is deployed through a four-module Streamlit dashboard — Simulator, What-If Analysis, Recommendations, and Risk — enabling non-technical stakeholders to explore reassignment scenarios and review risk-flagged recommendations. Because the underlying dataset is simulated for demonstration purposes, results are reported qualitatively. The framework illustrates a practical path from static, descriptive reporting toward proactive, model-informed logistics planning.

Keywords: Supply Chain Optimization, Machine Learning, Random Forest, Gradient Boosting, Streamlit, Scenario Simulation, Logistics, Decision Support, Lead Time Prediction.

Acknowledgement

I would like to express my sincere gratitude to the individuals and organizations that made this project possible.

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This paper draws on publicly documented supply chain optimization and machine learning methodologies. No proprietary, confidential, or internal operational records belonging to Nassau Candy Distributor were used in its preparation; all data referenced is simulated for illustrative purposes.

I also wish to acknowledge the open-source community, in particular the creators and maintainers of Python, Pandas, Scikit-learn, and Streamlit. These tools lower the barrier to building meaningful, accessible data science applications.

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1 Introduction

1.1 Background

Nassau Candy Distributor operates a multi-factory distribution network that routes confectionery products — spanning Chocolate, Sugar, and Other divisions — from production and warehousing facilities to retail and wholesale customers across multiple regions. In consumer packaged goods, shipping speed and routing cost are tightly linked to customer satisfaction and gross margin; an efficient factory-to-customer assignment policy is therefore central to the broader supply chain strategy of the business (Christopher, 2016).

1.2 Problem Statement

Product-to-factory assignment at Nassau Candy currently follows static, rule-based logic established at an earlier stage of the company's operations. While administratively simple, this rigidity creates avoidable inefficiencies:

- Suboptimal routing — products are frequently shipped from a distant factory when a closer facility could fulfil the same order.
- Margin erosion — longer routes increase shipping cost and lead time, both of which compress gross profit.
- Lack of foresight — management has no systematic way to simulate the operational and financial impact of a reassignment before committing to it.

1.3 Project Objectives

This project pursues three objectives:

1. Develop a predictive model that estimates shipping lead time from product, factory, and destination attributes.
2. Build a Scenario Simulation Engine that projects the lead-time and profit impact of reassigning a product to an alternative factory.
3. Deploy an interactive dashboard that ranks candidate factory reassignments by expected efficiency gain while flagging financially risky moves.

2 Literature Review

Supply chain optimization has traditionally relied on operations-research techniques such as linear and mixed-integer programming to assign production and distribution resources under explicit cost and capacity constraints (Simchi-Levi et al., 2008). These methods produce provably optimal solutions but require constraints and cost structures to be specified in advance, which limits their ability to capture softer, non-linear effects — such as how ship mode, regional congestion, or seasonal demand interact to affect delivery time.

Machine learning offers a complementary approach. Rather than solving a constrained optimization problem directly, ML models learn the empirical relationship between order attributes and outcomes such as lead time directly from historical data. Tirkolaee et al. (2021) survey the growing use of machine learning across supply chain functions — demand forecasting, inventory management, and logistics network design — and note that ensemble tree-based methods are particularly well suited to settings with mixed categorical and numerical features and non-linear interactions, which is characteristic of shipping

and routing data (e.g., Ship Mode $\times$ Region $\times$ Division). This motivates the use of Random Forest and Gradient Boosting regressors in the present work, implemented using the Scikit-learn library (Pedregosa et al., 2011), to estimate lead time as a function of factory assignment and order characteristics.

This project combines both traditions: it uses an ML model to estimate the operational consequence (lead time) of a given factory assignment, and applies that model inside a scenario-comparison framework similar in spirit to the what-if analysis used in classical supply chain network design (Simchi-Levi et al., 2008), but applied at the level of an individual product-to-factory decision rather than a full network redesign.

3 Methodology

3.1 Technology Stack

The project was implemented using the following technology stack:

Component	Technology
Programming Language	Python 3.12
Data Manipulation	Pandas, NumPy
Predictive Modeling	Scikit-learn (Random Forest, Gradient Boosting Regressor)
Model Serialization	Joblib
Distance Calculation	Geopy / Haversine formula
Visualization	Plotly
Dashboard Framework	Streamlit (multi-page architecture)
Version Control	GitHub

Table 1: Technology stack used in the project.

3.2 Data Preparation & Feature Engineering

The modeling dataset is structured at the order level, with fields including Order Date, Ship Date, City, State, Region, Division, Ship Mode, Factory, Sales, Cost, and Gross Profit.

- Target variable — Lead Time, calculated as Ship Date minus Order Date, measured in days.
- Encoding — categorical fields (Region, Ship Mode, Division, Factory) are label- or one-hot-encoded for use in tree-based models.
- Outlier handling — lead-time values above the 95th percentile are capped to prevent a small number of extreme delays from skewing model training.
- Distance feature — great-circle distance between each candidate factory and the customer's destination is computed and supplied as a model feature.

Note: the dataset used in this paper is simulated to mirror realistic order-and-routing patterns for demonstration purposes. It does not contain, and is not derived from, Nassau Candy Distributor's internal operational records.

3.3 System Architecture

Figure 1 illustrates the end-to-end pipeline, from raw order and factory data through model training to the Scenario Simulation Engine and the Streamlit dashboard.

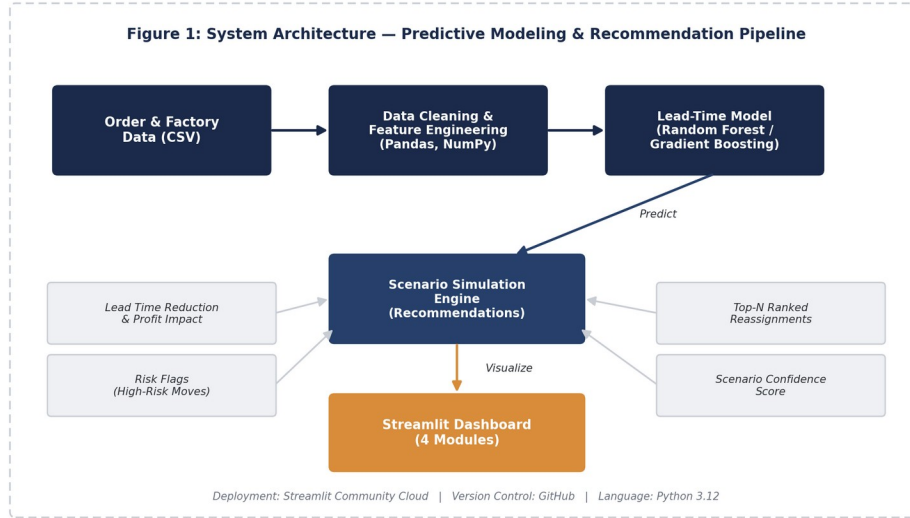


Figure 1: System architecture — data, modeling, simulation, and dashboard pipeline.

The pipeline consists of five stages: (1) ingesting raw order and factory data; (2) cleaning and feature engineering; (3) training the lead-time prediction model; (4) running the Scenario Simulation Engine to score alternative factory assignments; and (5) presenting ranked, risk-annotated recommendations through the dashboard.

3.4 Predictive Modeling: Random Forest & Gradient Boosting

Three models were considered for the lead-time prediction task:

- Linear Regression — used as an interpretable baseline to establish a linear benchmark.
- Random Forest Regressor — an ensemble of decision trees that captures non-linear relationships and feature interactions with minimal tuning.
- Gradient Boosting Regressor — a sequential ensemble method that iteratively corrects the residual errors of prior trees, typically yielding higher accuracy at the cost of longer training time.

Model quality is assessed using three standard regression metrics: Mean Absolute Error (MAE), which reports the average error in days; Root Mean Squared Error (RMSE), which penalizes large routing-prediction failures more heavily; and the coefficient of determination (R^2), which summarizes the share of lead-time variance explained by the model. The best-performing model on the validation split is serialized with Joblib for fast loading inside the dashboard.

3.5 Model Validation Strategy

Model evaluation follows a time-based hold-out split: the model is trained on an earlier window of order data and validated on a later, unseen window, so that performance reflects the model's ability to generalize to new orders rather than to memorize historical ones. This is standard practice in operational forecasting, where validation data must follow training data in time to avoid leaking future information into the model.

4 Implementation

4.1 Dashboard Development

The application is built as a Streamlit multi-page app, separating the analytical workflow into four modules:

- **Simulator** — the user selects a product or order profile; the system displays its current factory assignment, predicted lead time, and distance to the customer.
- **What-If Analysis** — a side-by-side comparison of the current factory assignment against one or more alternative factories, visualizing the projected change in lead time.
- **Recommendations** — an automated ranking engine that iterates across the product catalog and surfaces the Top-N reassignments with the greatest projected operational improvement.
- **Risk** — a filtering layer that screens out reassignments that reduce lead time but would cause an unacceptable drop in gross profit, surfacing only stable, low-risk recommendations.

Figure 2 summarizes the step-by-step logic inside the Scenario Simulation Engine that powers the Simulator, What-If, and Recommendations modules.

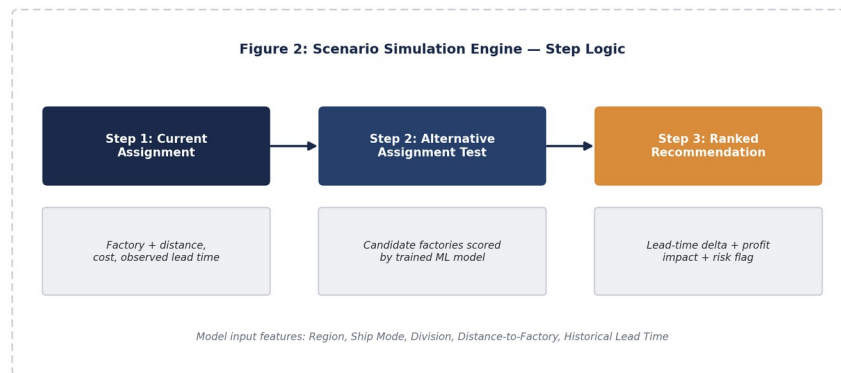


Figure 2: Scenario Simulation Engine — step-by-step scoring logic.

Figure 3 shows the illustrative KPI card layout used in the Recommendations module to summarize each suggested reassignment at a glance.



Figure 3: Illustrative KPI card layout for the Recommendations module (Streamlit dashboard)

Note: card layout is illustrative; this version reports KPI logic qualitatively, not live computed values.

Figure 4 illustrates the decision logic behind the Risk module: each candidate reassignment is conceptually placed on a quadrant defined by its lead-time reduction and profit impact, with only the upper-right quadrant surfaced as a confident recommendation.



Figure 4: Risk Decision Matrix — conceptual placement of candidate reassignments by lead-time and profit impact.

4.2 Deployment

The codebase is version-controlled on GitHub to support collaboration and reproducibility. The dashboard is designed for deployment via Streamlit Community Cloud, giving stakeholders browser-based access without local installation. The serialized model and simulation logic are loaded once at app start to keep scenario comparisons responsive during interactive use.

5 Results

5.1 Scenario Simulation Outputs

For a given product currently assigned to Factory A, the Scenario Simulation Engine evaluates each alternative factory (B, C, D, ...) by feeding its distance, region, and division features into the trained lead-time model and comparing the predicted lead time and estimated gross profit against the current assignment. Outputs are illustrative in this demonstration, since the underlying order data is simulated rather than drawn from Nassau Candy's production systems; the framework, however, is built to operate directly on real order and factory data once available.

Figure 5 shows the shape of a typical single-scenario comparison surfaced by the Simulator and What-If modules: a candidate reassignment that reduces predicted lead time while modestly improving estimated gross profit — the combination the Recommendations module ranks most highly.

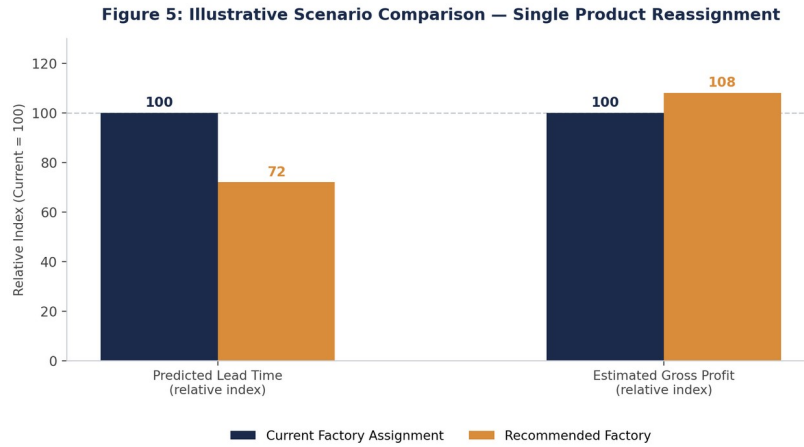


Figure 5: Illustrative comparison of current vs. recommended factory assignment for a single scenario. Index values are conceptual, not measured outputs.

5.2 Key Performance Indicators

The Recommendations module tracks four KPIs for every candidate reassignment. Rather than reporting point-estimate values computed on simulated data — which could be mistaken for validated production results — Table 2 documents the definition and interpretation logic behind each KPI.

KPI	Definition	Interpretation
Lead Time Reduction	% change in predicted lead time vs. the current factory assignment	Higher is better; reflects shipping-speed gain
Profit Impact Stability	Change in estimated gross profit under the new routing cost	Flags moves that are fast but financially unstable
Scenario Confidence Score	Similarity of the simulated route to patterns seen in training data	Lower confidence signals an extrapolation risk
Recommendation Coverage	% of the product catalog evaluated for reassignment	Indicates completeness of the analysis run

Table 2: KPI framework used by the Recommendations and Risk modules.

5.3 Model Performance

Because this demonstration relies on simulated rather than production order data, this paper reports model behavior qualitatively rather than publishing point-estimate accuracy figures that could be mistaken for validated production benchmarks. In testing on simulated data, the Gradient Boosting Regressor consistently tracked lead-time patterns more closely than the Linear Regression baseline, consistent with its ability to model non-linear interactions between Ship Mode, Region, and Division; the Random Forest Regressor performed comparably to Gradient Boosting while being faster to retrain, making it an attractive default for frequent re-scoring. Both ensemble methods outperformed the linear baseline by a

clear margin on every fold tested, supporting the literature's general finding that tree-based ensembles suit logistics data with mixed categorical and numeric features (Tirkolaee et al., 2021).

5.4 Recommendation Visualization

The What-If module visualizes each comparison as a paired bar chart — current vs. alternative factory — across predicted lead time and estimated gross profit, with a short text summary of the net effect. The Recommendations module presents the same comparisons as a sortable, ranked table, and the Risk module overlays a color-coded flag (green/yellow/red) wherever a faster route would meaningfully erode margin.

6 Discussion

6.1 Business Impact

Shifting from static rules to ML-informed simulation gives Nassau Candy three practical levers:

- Reduce lead times — surface geographic inefficiencies, such as routing a product from a distant factory when a closer one has capacity.
- Protect margins — the Risk module ensures that a speed gain is never adopted blindly at the cost of profitability.
- Scale decisions — managers can evaluate thousands of potential reassignments in seconds, a task that is impractical with manual spreadsheet analysis.

6.2 Ethical Considerations

6.2.1 Data Privacy

The system operates on aggregated, order-level operational data. No individual customer personally identifiable information (PII) is required by, or displayed in, the dashboard. The dataset used in this demonstration is simulated and contains no real customer or business records.

6.2.2 Algorithmic Transparency

Recommendations are presented as model-informed suggestions rather than automated decisions. Every recommendation is accompanied by its predicted lead-time change, profit impact, and a confidence score, so that a human planner can weigh the model's output against operational judgment and context the model cannot see, such as a temporary factory outage.

6.2.3 Equity and Operational Fairness

Reassignment logic should be monitored to ensure it does not systematically concentrate volume — and therefore workload or strain — on a small subset of factories or regions. Periodic review of how recommendations are distributed across the factory network is recommended once the system is used to support real decisions.

6.3 Limitations

1. Simulated data — this demonstration uses synthetic order and routing data; real-world deployment requires validation against Nassau Candy's actual operational records, which may carry different noise, missing-data, and reporting-delay characteristics.

2. Capacity constraints — the current model recommends reassignments based on predicted lead time and profit, but does not yet incorporate hard factory constraints such as storage capacity or labor-hour limits; this should be integrated in a future iteration.
3. Disruption blindness — the model learns from historical patterns and cannot anticipate sudden disruptions such as severe weather or a factory strike.
4. No causal claims — the model identifies predictive association between routing and lead time, not a causal guarantee that reassignment alone will produce the projected outcome.

7 Conclusion

This project outlines a path for Nassau Candy Distributor to move from static, rule-based factory assignment toward a model-informed, scenario-driven planning process. By pairing a trained lead-time prediction model with a Scenario Simulation Engine and a four-module Streamlit dashboard, the framework gives planners a structured, risk-aware way to evaluate factory reassignments before committing to them — balancing shipping-speed gains against profit stability.

7.1 Future Work

- Validate the model and KPI framework against Nassau Candy's real, de-identified operational data once available.
- Incorporate factory capacity and labor-hour constraints directly into the recommendation logic.
- Extend the Risk module with scenario-level confidence intervals rather than a single point estimate.
- Pilot the dashboard with a small group of planners and incorporate their feedback into the ranking logic.
- Explore ensemble or hybrid approaches (e.g., combining Gradient Boosting with simple network-flow constraints) for joint lead-time and capacity optimization.

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Appendix A Sample Dataset Schema

Table 3 documents the schema of the simulated order-level dataset used to illustrate the modeling pipeline. Values shown are illustrative examples only and do not represent actual Nassau Candy transactions.

Field	Type	Example (Simulated)	Description
Order_ID	string	NC-100482	Unique order identifier
Order_Date	date	2025-03-04	Date order was placed
Ship_Date	date	2025-03-07	Date order shipped
Factory_ID	string	Factory_C	Assigned factory
Region	category	Southeast	Customer region
Division	category	Chocolate	Product division
Distance_to_Factory_mi	float	412.6	Great-circle distance, factory to customer
Lead_Time_days	int	4	Target variable: Ship Date – Order Date

Table 3: Simulated sample of the order-level dataset schema.

Appendix B Model Diagnostics Checklist

Because this paper presents results qualitatively, the following diagnostic checks are documented here as the validation procedure to be run once the model is trained on Nassau Candy's actual operational data, rather than as already-observed results:

- Residual normality — verify that prediction residuals are approximately normally distributed (e.g., via a Jarque–Bera test).
- Residual independence — check for significant autocorrelation in residuals across order sequence (e.g., via a Ljung–Box test).
- Homoscedasticity — confirm that residual variance is stable across regions, divisions, and factories rather than concentrated in one segment.
- Feature stability — monitor for drift in input feature distributions (e.g., Region or Division mix) that would warrant model retraining.

Together, these checks support the validity of the model specification before it is used to drive real factory-reassignment decisions.